Physics Important Facts

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Introduction

- **Nature of the Universe**
- Basic structure, celestial bodies, galaxies, universe expansion.
- **General Scientific Laws**
- Newton's Laws, Conservation Laws, Ohm's Law, Boyle's Law, etc.
- **Inventions and Discoveries in Physics**
- Milestone discoveries, inventors and their contributions.
- **Nobel Prize in Physics**
- Notable Indian and global Nobel laureates and their work.
- **National Scientific Laboratories**
- ISRO, DRDO, BARC, CSIR, IITs – their functions and achievements.
- **Physical Quantities, Standards and Units**
- SI Units, Derived Units, Measurement devices.
- **Mechanics and Properties of Matter**
- Motion, gravity, elasticity, viscosity, pressure, etc.
- **Force, Motion and Energy**
- Types of force, work, energy, power, laws of motion.
- **Energy**
- Forms of energy, renewable and non-renewable sources, solar, wind, hydro.
- **Heat, Light and Sound**
- Temperature, reflection, refraction, echo, pitch, dispersion, etc.

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DEPARTMENT OF EMPLOYMENT AND TRAINING
UNIT I: PHYSICS MINDMAP

NATURE OF UNIVERSE

UNIVERSE
- The Universe is commonly defined as the totality of existence, including planets, stars and galaxies, the contents of intergalactic space, and all matter and energy

Galaxy
- A galaxy is a collection of stars, dust and gas. All these materials bound together gravitationally to form a galaxy

Star
- A star is a huge, shining ball that produces a large amount of energy in form of light and other forms

Constellation
- The stars forming a group that has a recognizable shape is called a Constellation

ASTEROIDS

Asteroids are small rocky bodies that orbit the sun in a belt (Kuiper belt) between orbits of Mars and Jupiter

METEORS

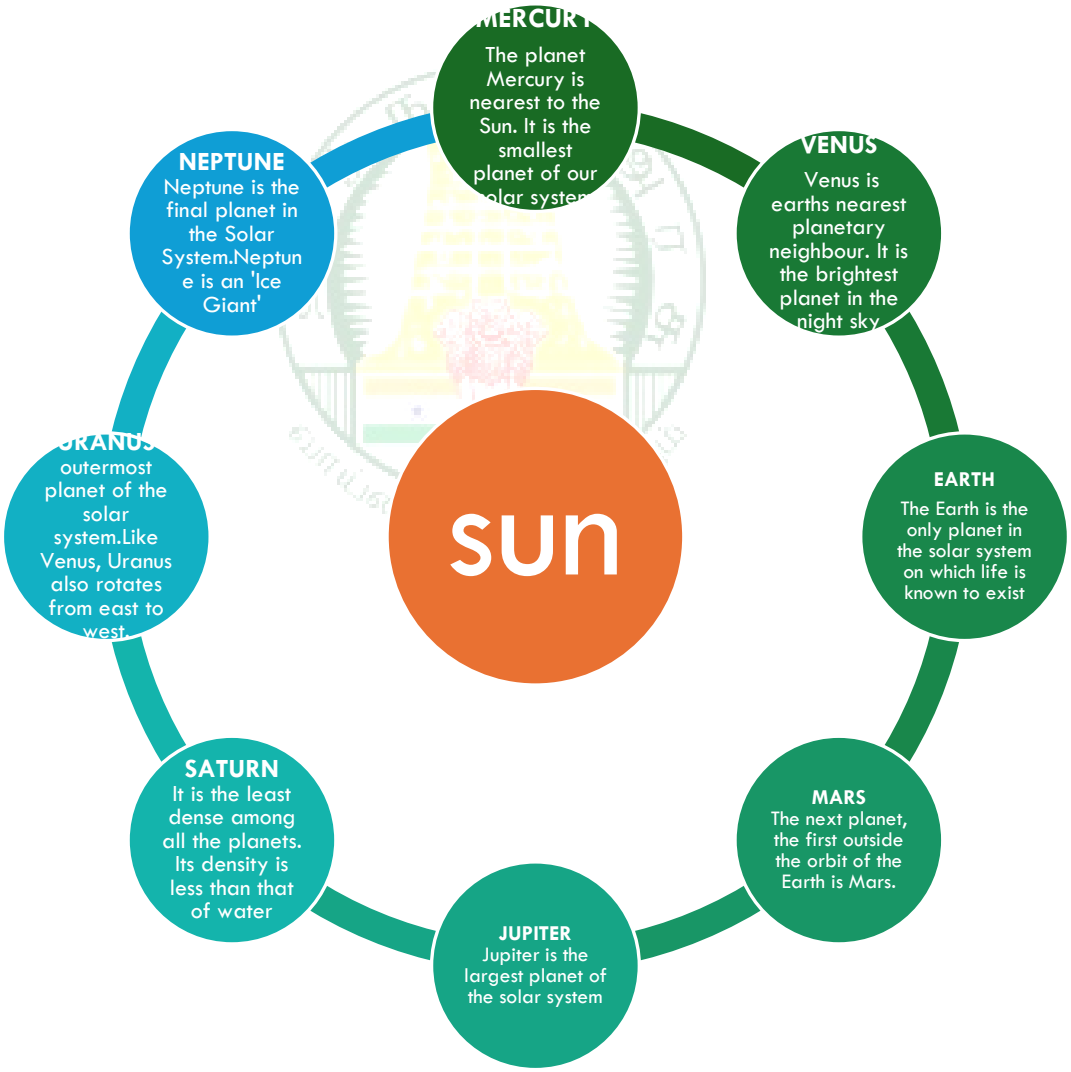
A meteor is a bright streak of light that appears briefly in the sky. Meteors are often called shooting stars or falling stars because they look like stars falling from the sky

COMETS

Comets revolve around the Sun in highly elliptical orbits. However, their period of revolution around the Sun is usually very long. A Comet appears generally as a bright head with a long tail. The length of the tail grows in size as it approaches the Sun. The tail of a comet is always directed away from the Sun.

THE SOLAR SYSTEM

The Sun and the celestial bodies which revolve around it form the solar system.



Gravitational Force

We are all aware of the force which pulls us towards the earth. This is called gravitational force. In fact the gravitational force exists between all bodies, even between two apples lying on a table

Newton's Law of Universal Gravitation

F = Gm1m2 / r2 where r is the distance between two particles of masses m1 and m2 and G the universal gravitational constant. The value of G is 6.67 x 10 –11 SI units

Centripetal Force & Centrifugal Force

Centripetal Force

For a body to move in a circle there must be a force on it directed towards the centre. This is called the centripetal force and is necessary to produce continuous change of direction in a circular motion .Fc = mv2 / r

Centrifugal Force

This force is supposed to be acting on a body revolving in a circle. Centrifugal force is equal and opposite to centripetal force, i.e., it acts outwards.

Weight

The weight of a body is the force with which the earth attracts the body towards its centre. The weight of a body can also be expressed in terms of the acceleration due to gravity (g). The weight (W) of a body of mass m is given by the product of m and g, i.e., W = mg

GENERAL SCIENTIFIC LAWS

Newton's Laws of Motion

Newton's three laws of motion sum up the basic principles of motion

First Law

- Every object continues in its state of rest or of uniform motion in a straight line if no net force acts upon it

Second Law

- This law states that "the rate of change of momentum of a body is proportional to the applied force and takes place in the direction of the force."
- F = ma

Third Law

- This law states that "to every action there is an equal and opposite reaction."

MOMENTUM

The momentum of a body is defined as the product of its mass and velocity.

Momentum = mass x velocity

Conservation of Energy

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Energy can neither be created nor destroyed; it can only be transformed from one form into another, but the total amount of energy remains constant

Newton's Law of Universal Gravitation

- Newton's Law of Universal Gravitation states that every particle in the universe attracts every other particle with a force that is directly proportional to the product of their masses and inversely proportional to the square of the distance between them.
- In equation form, the gravitational force, F = Gm1m2 / r2 where r is the distance between two particles of masses m1 and m2 and G the universal gravitational constant. The value of G is 6.67 x 10 –11 SI units

Archimedes' Principle

This principle states that when a body is wholly or partially immersed in a fluid, it experiences an upthrust equal to the weight of the fluid displaced.

When an object is immersed in a fluid, two forces act on it:

(i)) the weight of the object acting downward,

(ii) upthrust acting upward.

It is due to upthrust that objects apparently weigh less when immersed in fluids.

Kirchhoff's rule

- According to the Junction rule in a circuit, the total of the currents in a junction is equal to the sum of currents outside the junction.
- According to the loop rule, the sum of the voltages around the closed loop is equal to null.

Kirchhoff's First Law

According to Kirchhoff's Current law, the total current entering a junction or a node is equal to the charge leaving the node as no charge is lost. Put differently, the algebraic sum of every current entering and leaving the node has to be null.

Kirchhoff's Second Law

According to Kirchhoff's voltage law, the voltage around a loop equals to the sum of every voltage drop in the same loop for any closed network and also equals to zero. Put differently, the algebraic sum of every voltage in the loop has to be equal to zero and this property of Kirchhoff's law is called as conservation of energy.

Pascal's Law

According to this principle, the external static pressure applied on a confined liquid is distributed or transmitted evenly throughout the liquid in all directions.

Pascal's Law formula.

$$F = PA$$ Where,

F is the force applied,

P is the pressure transmitted,

A is the cross sectional area

Applications of Pascal's Law

Hydraulic Lift: It works based on the principle of equal pressure transmission throughout a fluid (Pascal's Law).

Inventions and Discoveries

Year	Discoveries	Scientist Name
1905	Photon	**Albert Einstein**
1897	Electron	**J.J.Thomson**
1919	Proton	**Ernest Rutherford**
1808	Atom	**John Dalton**
1932	Neutron	**James Chadwick**
1687	Law of Motion	**Isaac Newton**
1779	Coulomb	**Charles-Augustinde Coulomb**
1827	Ohm'sLaw	**Georg Simon Ohm**
1831	Electromagnetic Induction	**Michael Faraday**
1880	Thermionic Emission	**Thomas Edison**
1896	Radio activity	**Henri Becquerel**
1898	Radium	**Marie Sklodowska-Curie**
1900	Quantum theory	**Max Planck**
1905	Photoelectric Effect	**Heinrich Rudolf Hertz & Albert Einstein**
1895	X-Ray	**Röntgen**
1905	Relativity	**Albert Einstein**
1913	Atomic Structure	**Neil Bohr & Rutherford**
1942	Nuclear Reactor	**Anrico Fermi**
1904	Diode Bulb	**Sir J.S Fleming**

1906	Triode Bulb	**Leede Forest**
1608	**Telescope**	**Hann Lippershey**

Invention	Inventor
Centigrade scale	Anders elsius
Watch	Peter Henlein
Radio	Guglielmo Marconi
Telephone	Alexander Graham Bell
Electricity	Benjamin Franklin
Electric Light Bulb	Thomas Edison
Thermometer	Galileo Galilei
Telescope	HansLippersheyandZachariasJanssen; later Galileo
Telegraph	Samuel Morse
Cosmic Rays	Victor Hess (buttheterm‘cosmicrays’first used by Robert Millikan
Automobile	Karl Benz
MagneticTape	Fritz Pfleumer
Transformer	Michael Faraday(laterOttóTituszBláthy)
Electromagnetic Induction	Michael Faraday
Quantum mechanics	Werner Heisenberg, Max Born,and Pascual Jordan
Wave mechanics	Erwin Schrödinger
Fuel Cell	William Grove
Airplane	Wright Brothers
Barometer	Evangelista Torricelli
Camera	Nicéphore Niépce
Diesel Engine	Rudolf Diesel
Helicopter	Igor Sikorsky

Dynamite	**Alfred Nobel**

Lift	**Elisha Otis**
Laser Printer	**Gary Starkweather**
Mobile Phone	**Martin Cooper**
Printing Press	**Johannes Gutenberg**
Video Games	**Ralph Baer**
Steam engine	**Thomas Newcomen**
Railway Engine	**George Stephenson**
Jet Engine	**Frank Whittle**
Seismograph	**John Milne**
Electric Generator	**Michael Faraday**
Television	**John Logie Baird**
Refrigerator	**William Cullen (laterOliverEvans)**
Carburetor	**Luigi De Cristoforis & Enrico Bernardi**
Air Brake	**George Westing house**
Atomic bomb	**Robert Oppen heimer,Edward Telleretal**
Airconditioner	**Willis Carrier**
Machine Gun	**Sir Hiram Maxim**
Radar	**SirRobertAlexanderWatson-Watt**
Submarine	**CorneliusDrebbel(later)DavidBushnell**
First military submarine	**Yefim Nikonov**
Transistor	**JohnBardeen,WalterBrattain,andWilliam Shockley**
Galvanometer	**Johann Schweigger**
Laser	**Theodore H.Maiman (firstdemonstrated)**
Neon lamp	**Georges Claude**
Rocket Engine	**Robert Goddard**

Typewriter	**Christopher LathamSholes**

Nobel Prize in Physics

- The Nobel Prize in Physics is the most prestigious award given yearly by the Royal Swedish Academy of Sciences.
- The Noble prize is given to those physicists who conferred the most outstanding contributions for mankind (in physics).
- Wilhelm Röntgen, a German/Dutch physicist, was the first person who had received the first Nobel Prize in 1901.
- Wilhelm Röntgen had received the Nobel Prize for discovery of the remarkable x-rays).
- In the field of physics (by the time), only two women have won the Nobel Prize, namely Marie Curie (in 1903) and Maria Goeppert Mayer (in 1963).
- The following table illustrates some of the significant physicists who have received the Nobel Prize along with their remarkable works

Name	Year:Country	Work
Wilhelm Conrad Röntgen	1901:Germany	**Discovery of the remarkable rays**
Hendrik Lorentz	1902:Netherlands	**Worked on the influence of magnetism upon radiation phenomena**
Pieter Zeeman		
Antoine Henri Becquerel	1903:France	**Spontaneous radioactivity**
Pierre Curie		**Radiation phenomena**
Maria Skłodowska-Curie	1903: Poland/France	
Philipp Eduard Antonvon Lenard	1905:Austria - Hungary	**Worked on cathode rays**
Guglielmo Marconi	1909:Italy	**Development of wireless telegraphy**
Karl Ferdinand	**1909:Germany**	

Braun		
Max Planck	1918:Germany	**Discovered energy quanta**
Johannes Stark	1919:Germany	**Discovered Doppler effectincanal rays**
AlbertEinstein	1921:German y- Switzerland	**For the discovery of the law of the photoelectric effect**
Niels Bohr	1922:Denmark	**Investigated the structure of atoms**
Chandrasekhara Venkata Raman	1930:India	**Worked on scattering of light**
Werner Heisenberg	1932:Germany	**Created quantum mechanics**
ErwinSchrödinger	1933:Austria	**Discovered productive forms of atomic theory**
Paul Dirac	1933:United Kingdom	
James Chadwick	1935: UK	**Discovered Neutron**
Victor Francis Hess	1936:Austria	**Discovered cosmic radiation**
Willis EugeneLamb	1955: US	**Discovered the fine structure of the hydrogen spectrum**
Emilio Gino Segrè	1959:Italy	**Discovered the antiproton**
Owen Chamberlain	1959: US	
Lev Davidovich Landau	1962:Soviet Union	**Theories for condensed matter**
Maria Goeppert-Mayer	1963: US	**Discovered nuclear shell structure**
J.Hans D.Jensen	1963:Germany	
Hans	1967: US	**Worked on the theory of nuclear**

AlbrechtBethe		**reactions**
Murray Gell-Mann	1969: US	**Classification of elementary particles and their interaction**
Hannes Ol of Gösta Alfvén	**1970:Sweden**	**Worked on plasma physics**
Louis Néel	**1970:France**	**Worked solid state physics (anti ferro magnetism and ferrimagnetism)**
Dennis Gabor	1971:Hungary- UK	**Developed the holographic method**
John Bardeen	1972: US	**Developed the theory of superconductivity**
Leon Neil Cooper		
John Robert Schrieffer		
Arno Allan Penzias	1978: US	**Discovered cosmic microwave background radiation**
Robert Woodrow Wilson		
Nicolaas Bloembergen	1981: Netherlands-US	**Developed laser spectroscopy**
Arthur Leonard Schawlow	1981: US	
Ernst Ruska	1986:Germany	**Designed the first electron microscope**
Johannes Georg Bednorz	1987:Germany	**Discovered the superconductivity in**

Karl Alexander Müller	1987:Switzerland	**ceramic materials**
Robert B. Laughlin	1998: US	**Discovered a new form of quantumfluid**
Horst Ludwig Störmer	1998:Germany	
Daniel Chee Tsui	1998:China-US	
Jack St.Clair Kilby	2000: US	**Developed integrated circuit**
Riccardo Giacconi	2002:Italy-US	**Discovered cosmicX-ra**
Willard S.Boyle	**2009:Canada-US**	**Invented an imaging semiconductor**
George E. Smith	2009: US	**Worked on the quantum theory of circuit – the CCD sensor optical coherence**
Takaaki Kajita	2015:Japan	**Discovered neutrino oscillations, which illustrations that the neutrino shave mass**
Arthur B.Mc Donald	2015:Canada	
J.Michael Kosterlitz	2016: UK	**Use of topological concept sinphysics and discovered that superconductivity could occur at low temperatures**
DuncanHaldane	2016: London	
David J.Thouless	2016: US	
Rainer Weiss	2017: US	**Laser Inter ferometer Gravitational-Wave Observatory, or Ligo**
Kip S. Thorne	2017: US	
Barry C.Barish	2017:US	
Arthur Ashkin	2018: US	**Opticaltweezersandtheirapplicationto biological systems**
Gérard Mourou	2018:France	**In the field of laser physics for inventing the method of generating high-intensity, ultra- short optical pulses**
Donna Strickland	**2018:Canada-US**	

National Scientific Laboratories

Institute	Location
All India Institute of Medical Sciences (AIIMS)	New Delhi, Bhopal, Bhubaneswar, Jodhpur,Patna,Raipur,Rishikesh,etc.
All India Institute of Speech and Hearing (AIISH)	Mysore
Aryabhatta Research Institute of Observational Sciences (ARIES)	Nainital
Birbal Sahni Institute of Palaeobotany	Lucknow
Bose Institute	Kolkata
Central Drug Research Institute	Lucknow
Centre for the Study of Developing Societies (CSDS)	Delhi
Central Electronics Engineering Research Institute	Pilani
Central Food Technological Research Institute	Mysore
Central Glass and Ceramic Research Institute (CGCRI)	Kolkata
Central Institute for Cotton Research	Nagpur
Central Institute of Agricultural Engineering	Bhopal
Central Institute of Brackish Water Aquaculture	Chennai
Central Institute of Educational Technology	NewDelhi
Central Institute of Fisheries Education	Mumbai
Central Institute of Fisheries,Nautical and	Cochin

Institute	Location
Engineering Training(CIFNET)	
Central Institute of Fresh Water Aquaculture	Bhubaneswar
Central Institute of Indian Languages	Mysore
Central Institute of Plastics Engineering &Technology	Chennai
Central Institute of Psychiatry	Ranchi
Central Institute of Medicinal and Aromatic Plants	Lucknow
Central Institute of Road Transport(CIRT)	Pune
Central Leather Research Institute	Adyar,Chennai
Central Institute of Technology,Kokrajhar (CIT)	Kokrajhar
Central Mine Planning and Design Institute Limited	Ranchi
Central Research Institute for Dryland Agriculture	Hyderabad
Central Sheep and Wool Research Institute	Avikanagar
Central Silk Board	Bangalore, Mysore, Ranchi, Berhampore, Jorhat, Pampore, Guwahati,Delhi,Mumbai,Kolkata, Chennai, Hyderabad
Central Soil Salinity Research Institute	NewDelhi
Centre for Cultural Resources and Training (CCRT)	NewDelhi
Centre for Development Studies	Thiruvanathapuram

Centre for Excellence in Basic Sciences	Mumbai

Institute	Location
College of Defence Management	Secunderabad
Film and Television Institute of India	Pune
Foreign Service Institute	NewDelhi
Harcourt Butler Technological Institute	Kanpur
Harish Chandra Research Institute(HRI)	Allahabad
Indian Agricultural Research Institute(IARI)	NewDelhi
Indian Association for the Cultivation of Science (IACS)	Kolkata
Indian Diamond Institute	Surat
Indian Institute of Architects	Mumbai
Indian Institute of Astrophysics	Bangalore
Indian Institute of Chemical Biology(IICB)	Kolkata
Indian Institute of Chemical Technology	Hyderabad
Indian Institute of Coal Management	Ranchi
Indian Institute of Ecology and Environment	NewDelhi
Indian Institute of Engineering Science and Technology (IIEST)	Shibpur
Indian Institute of Foreign Trade (IIFT)	NewDelhi,Kolkata
Indian Institute of Forest Management(IIFM)	Bhopal

Indian Institute of Information Technology (IIIT)	Gwalior,Allahabad,Ranchi,Dharwad, Jabalpur, Kanchipuram, Kottayam, Kurnool, Guwahati, Bangalore, Sri City
Indian Institute of Catering Technology and Hotel Management	Tanjore

Institute	Location
National Institute of Securities Markets (NISM)	Navi Mumbai
National Institute Of Technical Teachers Training and Research	Bhopal,Chandigarh,Chennaiand Kolkata
National Institute of Technology	Agartala,Aizawl,Allahabad,Bhopal, Dimapur Nagaland, Durgapur, Goa, Hamirpur, Imphal, Jaipur, Jalandhar, Jamshedpur, Karaikal, Calicut, Kurukshetra, Nagpur, New Delhi, Patna, Raipur, Sikkim, Rourkela, Shillong, Silchar, Srinagar, Surat, Surathkal, Trichy, Tadepalligudem, Warangal, Yupia
National Institute of Urban Affairs	NewDelhi
National Institute of WaterSports	Goa
National Museum Institute of the History of Art, Conservation and Museology (NMIHACM)	NewDelhi
National Power Training Institute	Faridabad
National Sugar Institute	Kanpur
National Tuberculosis Institute	Bangalore
National Institute of Homoeopathy(NIH)	Kolkata
Netaji Subhas National Institute of Sports (NSNIS)	Patiala
North Eastern Regional Institute of Science and Technology (NERIST)	Itanagar
Physical Research Laboratory(PRL)	Ahmedabad
PEC University of Technology(PEC)	Chandigarh

Institute	Location
Raman Research Institute(RRI)	Bangalore
Rajiv Gandhi Institute of Petroleum Technology	RaeBareli
Saha Institute of Nuclear Physics	Kolkata
Sanjay Gandhi Post graduate Institute of Medical Sciences	Lucknow
Sant Longowal Institute of Engineering and Technology(SLIET)	Sangrur
Satyajit Ray Film and Television Institute	Kolkata
S.N.Bose National Centre for Basic Sciences(SNBNCBS)	Kolkata
School of Planning and Architecture	Vijayawada,Bhopal,Delhi
Sher-i-Kashmir Institute of Medical Sciences	Srinagar
South Asian University	NewDelhi
State Institute of Public Administration and Rural Development	Agartala
Tata Institute of Fundamental Research (TIFR)	Mumbai,Hyderabad
Tata Institute of Social Sciences	Mumbai,Hyderabad,Guwahati, Tuljapur
Rajiv Gandhi National Institute of Intellectual Property Management Nagpur	Nagpur
V.V.Giri National Labour Institute	NewDelhi
Variable Energy Cyclotron Center	Kolkata

Mechanics and Properties of Matter

Mechanics can be defined as the behavior of the particle under any effects.

KINEMATICS

Kinematics deals with the motion of particles not the causes of the motion.

Distance is a scalar quantity representing the interval between two points. It is just the magnitude of the interval.

Displacement is a vector quantity and can be defined by using distance concept. It can be defined as distance between the initial point and final point of an object. It must be the shortest interval connecting the initial and final point that is a straight line.

Speed can be defined as "how fast something moves" or it can be explained more scientifically as "the distance covered in a unit of time". It is a scalar quantity

Velocity can be defined as "speed having direction". It is a vector quantity having both magnitude and direction. Velocity may also be defined as the "rate of change of displacement" whereas "the speed is rate of change of distance"

Acceleration can be defined as "change in velocity". This change can be in the magnitude (speed) of the velocity or the direction of the velocity

FORCE

Force is a vector quantity having both magnitude and direction. The change in position of the object due to the work done or the cause of motion is "force". The unit of force is Newton or $kg.m/s^2$

MATTER AND ITS PROPERTIES

Everything around us has mass and volume and they occupy space, and we called them as matter. It can be in four sate, like solid, liquid, gas and plasma.

The main properties of matter that are discussed in this unit are mass, volume, density, elasticity, inertia, etc.

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UNIT I: PHYSICS MINDMAP

Mass
Mass is the quantity of the matter in a substance. Units of mass can be gram (g) or kilogram (kg)

Volume
Volume is the space occupied by the matter. It is also common property of matter and does not help us in distinguishing them. Its unit used in SI system is m^3

Density of Matter
Density is the quantity of mass in a unit of volume. It is the distinguishing property of matter. Each matter has its own density. Representation of density is d; unit is g/cm^3

Inertia
Inertia is one of the properties of matter. It is the resistance of the matter to change its state of motion. An unbalanced force can only change the state of motion of the matter

Elasticity
The property of a material to regain its original state when the deforming force is removed is called elasticity

SURFACE TENSION

- Surface tension is a contractive tendency of the surface of a liquid that allows it to resist an external force.
- It is revealed, for example, in the floating of some objects on the surface of water, even though they are denser than water, and in the ability of some insects (e.g. water striders) to run on the water surface.
- Surface tension has the dimension of force per unit length, or of energy per unit area

Applications of Surface Tension

- Emulsion
- Formation of drops occurs when a mass of liquid is stretched

- Flotation of objects denser than water occurs when the object is non-wettable and its weight is small enough to be borne by the forces arising from surface tension.

Bernoulli's theorem

Streamline flow of a non-viscous and incompressible liquid, the sum of the pressure energy, kinetic energy and potential energy per unit mass is a constant

Application of Bernoulli's theorem

- Lift of an aircraft wing
- Blowing of roofs
- Bunsen burner
- Motion of two parallel boats

Buoyant Force

When a body floats or immerses in a liquid, the pressure on the bottom surface is more than that the pressure on the top surface. Due to the difference in pressure, an upward force acts on the body. This upward force is called upthrust or buoyant force. The buoyant force is equal to the weight of the liquid displaced.

Archimedes Principle

tamilnaducareers

When a body is immersed in fluid (liquid or gas) it experiences an apparent loss of weight which is equal to the weight of the fluid displaced

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Hydrometers

- The laws of floatation are made use of in the construction of hydrometers used for the determination of the relative densities of solids and liquids.
- There are two types of hydrometers.
  1.The constant immersion hydrometer
  2.The variable immersion hydrometer
- A common hydrometer used to test the purity of milk by noting its specific gravity is called a LACTOMETER.

Physical Quantities, Standards and Units

Measurement

- Measurement is the assignment of a number to a characteristic of an object or event which can be compared with other objects or events.
- It is defined as the determination of the size or magnitude of something.

Physical quantities

- Physical quantity is a quantity that can be measured.

- Physical quantities can be classified into two

Fundamental quantities
Quantities which cannot be expressed in terms of any other physical quantities are called fundamental quantities.

Example: Length, mass, time, temperature

Derived quantities
Quantities like area, volume and density can be expressed in terms of some other quantities. They are called derived quantities. Physical quantities have a numerical value (a number) and a unit of measurement.

UNIT

- A unit is the standard quantity with which unknown quantities are compared.
- It is defined as a specific magnitude of a physical quantity that has been adopted by law or convention.
- For example, feet is the unit for measuring length

SI system of units

- SI system of units is the modernised and improved form of the previous system of units
- It is based on a certain set of fundamental units from which derived units are obtained by multiplication or division.
- The units used to measure the fundamental quantities are called fundamental units
- The units which are used to measure derived quantities are called derived units.
- There are seven fundamental units in the SI system of units. They are also known as base units.

Fundamental Physical Quantities and their units:

Fundamental quantities	Unit	Symbol
Length	metre	**m**
Mass	kilogram	**kg**

Time	second	**s**
Temperature	kelvin	**K**
Electric current	ampere	**A**
Luminous intensity	candela	**cd**
Amount of substance	**Mole**	**mol**

Derived Quantities and their units:

S.No	Physical quantity	Expression	Unit
1	Area	length×breadth	**m^2**
2	Volume	area×height	**m3**
3	Density	mass/volume	**Kgm^{-3}**
4	Velocity	displacement/time	**ms^{-1}**
5	Momentum	mass× velocity	**$Kgms^{-1}$**
6	Acceleration	velocity/time	**ms^{-2}**
7	Force	mass×acceleration	**$Kgms^{-2}$orN**
8	Pressure	force/area	**Nm^{-2}orPa**
9	Energy(work)	force×distance	**NmorJ**
10	**Surface tension**	**force/length**	**Nm-1**

Atomus:

- The smallest amount of time imaginable to us is a twinkling of the eye.
- This is called atomus. It is 1/6.25 seconds or 160 milliseconds.

Donkey Power:

- It is one third of a horse power.
- Its value is around 250watt

Some of the Measurements and units

- 1 AU is equal to 14,95,97,871 km or approximately equal to 150 million km or 1,500 lakhs km.
- 1 TMC is (thousand million cubic feet) hundred crore cubic feet.
- 1 TMC = 2.83 × 1010 litre.
- 1 TMC is approximately 3000 crore litres.
- The total length of all the blood vessels in human body is 96,000 km
- Light travels 3 × 108 m in one second or 3 lakhs kilometre in one second.
- In one year we have 365 days. The total number of seconds in one year is equal to 365 × 24 × 60 × 60 = 3.153 × 107second.
- 1 light year = (3.153×107) ×(3×108) = 9.46 × 1015 m.
- 1 parsec = 3.26 light year.
- 1 quintal = 100 kg
- 1 metric tonne = 1000 kg = 10 quintal

Atomic mass unit (amu):

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Mass of a proton, neutron and electron can be determined using atomic mass unit.
1 amu = 1/12th of the mass of carbon-12 atom

Temperature

- Temperature is the measure of hotness.
- SI unit of temperature is kelvin(K).
- Zero kelvin (0 K) is commonly known as absolute zero.
- C/100 = (F – 32)/180 = (K-273) /100

Units	Fahrenheit	Celsius	Kelvin
Fahrenheit(°F)	F	(F– 32) ×5/9	**(F– 32) ×5/9+273**
Celsius(°C)	(C×9/5)+32	C	**C+273**
Kelvin(K)	**(K–273) ×9/5+32**	**-273K**	**K**

FORCE, MOTION AND ENERGY

Force

Force is an external effort in the form of push or pull, which:

1. produces or tries to produce the motion of a static body.
2. stops or tries to stop a moving body.
3. changes or tries to change the direction of motion of a moving body.

Force has both magnitude and direction. So, it is a vector quantity

Types of forces

Like parallel forces: Two or more forces of equal or unequal magnitude acting along the same direction, parallel to each other are called like parallel forces

Unlike parallel forces: If two or more equal forces or unequal forces act along opposite directions parallel to each other, then they are called unlike parallel forces

Moment of the Force

- The rotating or turning effect of a force about a fixed point or fixed axis is called moment of the force about that point or torque (T).
- It is measured by the product of the force (F) and the perpendicular distance (d) between the fixed point or the fixed axis and the line of action of the force.

$$\tau = F \times d$$

- Torque is a vector quantity. It is acting along the direction, perpendicular to the plane containing the line of action.

Couple:

- Two equal and unlike parallel forces applied simultaneously at two distinct points
- constitute a couple
- Rotating effect of a couple is known as moment of a couple

Application of Torque



Inertia

- Inertia is one of the properties of matter. It is the resistance of the matter to change its state of motion. An unbalanced force can only change the state of motion of the matter.
- The inertia is of three types

Inertia of rest
- **Ex-** A person standing in a bus falls backward when the bus suddenly starts moving.

Inertia of motion
- **Ex-** When a passenger gets down from a moving bus, he falls down in the direction of the motion of the bus.

Inertia of direction
- **Ex-** When a bus moving along a straight line takes a turn to the right, the passengers are thrown towards left.

MOMENTUM

- The momentum of a body is defined as the product of its mass and velocity.
- Momentum = mass x velocity

Distance and Displacement

	Distance	Displacement
1.	Distance is the length of the actual path followed by an object or body while moving from one point to another.	Shortest distance between two points.
2.	Scalar quantity	Vector quantity
3.	Measure in meter in the SI system	Measure in meter in the SI system
4.	Not a unique quantity	Unique

5.	Distance can either be equal to or Greater than displacement	Displacement is either equal to Or less than the distance.

Speed and Velocity

	Speed	Velocity
1.	Change of distance with Respect to time	Change of displacement with Respect to time
2.	Scalar Quantity	Vector Quantity
3.	Measured in m/s in the S.I. System	Measured in m/s in the S.I. System
4.	Positive Quantity	Negative Quantity

ACCELERATION

- Acceleration can be defined as "change in velocity"
- Acceleration = change in velocity / time
- The unit of acceleration is m/s2 or ms-2

ENERGY

Energy can be defined as the capacity to do work

The mechanical energy of a body consists of potential energy and kinetic energy

Potential energy

- The potential energy of a body is the energy stored in the body by virtue of its position or the state of strain.
- EP= mgh

Kinetic energy

- The kinetic energy of a body is the energy possessed by the body by virtue of its motion.
- Kinetic energy Ek =½ mv2

Work

Work = Force x Distance

The Unit of Work is Nm

Power.

Power =Work Done / time

Its unit is watt and dimensional formula is ML2 T–3.

Conservation Of Energy

Nothing can be destroyed or created in the universe like energy

$V = \sqrt{2gh}$

Nuclear Energy

- Nuclear energy originates from the splitting of uranium atoms– a process called fission and fusion reactions

Nuclear Fusion

- In nuclear fusion, two low mass nuclei come into very close contact with each other, so that the strong force fuses them

Nuclear Fisson

- Nuclear fission is the reverse process of fusion.It is therefore possible for energy to be released if a heavy nucleus breaks apart into two lighter ones.

Chain Reaction

- For certain of the heaviest nuclei which produce neutrons on fission, and which also easily absorb neutrons to initiate fission, a self-igniting type of neutron-initiated fission can be obtained, in a so-called chain reaction.

Atomic energy

- Atomic energy is energy carried by atoms

Particle	Mass(kg)	Charge(coulomb)
Proton	1.672×10^{-27}	$+1.6 \times 10^{-19}$
Neutron	1.675×10^{-27}	**Neutral**
Electron	9.108×10^{-31}	-1.6×10^{-19}

Magnetism, Electricity and Electronics

Magnet

- A compass needle is a small magnet. Its one end, which points towards north, is called a north pole, and the other end which points towards south is called a south pole

Field lines

- Field lines are used to represent a magnetic field

Magnetic flux

- Magnetic flux is the number of magnetic field lines passing through a given area.
- It is denoted by ϕ and its unit is weber (Wb)

Properties of magnetic lines of force

Magnetic lines of force are closed continuous curves, extending through the body of the magnet.

Magnetic lines of force start from the North Pole and end at the South Pole.

Magnetic lines of force never intersect.

They will be maximum at the poles than at the equator.

The tangent drawn at any point on the curved line gives the direction of magnetic field

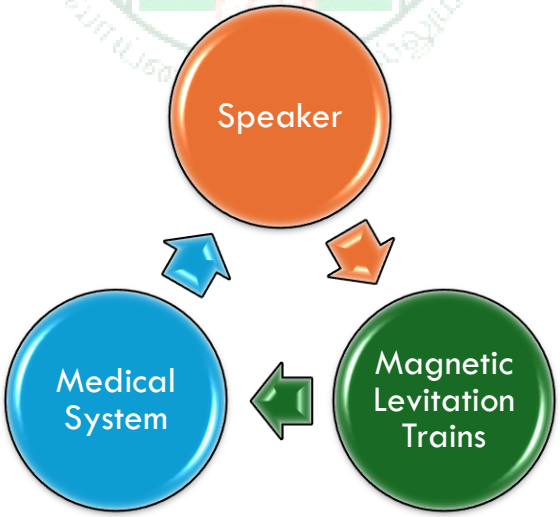
Diamagnetism, Paramagnetism and Ferromagnetism

Property	Diamagnetism	Paramagnetism	Ferromagnetism
Magnetic Behavior	Weakly repelled by a magnetic field	Weakly attracted to a magnetic field	Strongly attracted to a magnetic field
Electron Pairing	All electrons are paired	Some unpaired electrons	Many unpaired electrons
Magnetic Moment	No permanent magnetic moment	Small magnetic moment	Large, permanent magnetic moment
Retains Magnetism?	No	No	Yes (even after field is removed)
Examples	Bismuth, copper, silver	Aluminum, platinum, oxygen	Iron, cobalt, nickel

ELECTROMAGNET

- An electromagnet is a type of magnet whose magnetism is produced by the flow of electric current.

Applications of Electromagnets



ELECTRICITY

- A stream of electrons moving through a conductor constitutes an electric current

Electric force

- Among electric charges there are two types of electric force (F). One is attractive and another is repulsive.
- The like charges repel and unlike charges attract.
- The force existing between the charges is called as 'electric force'.

Electric current

- When the charged object is provided with a conducting path, electrons start to flow through the path from higher potential to lower potential region. Normally, the potential difference is produced by a cell or battery. When the electrons move, we say that an electric current is produced. That is, an electric current is formed by moving electrons.
- The SI unit of electric current is ampere. Ammeter is an instrument used to measure the strength of the electric current

OHM'S LAW

- The potential difference across the ends of a resistor is directly proportional to the current through it, provided its temperature remains the same.
- $V = IR$

Joule's Law of Heating

- Joule's law of heating states that the heat produced in any resistor is
  1. directly proportional to the resistance of the resistor.
  2. directly proportional to the square of the current passing through the resistor.
  3. directly proportional to the time for which the current is passing through the resistor

ELECTRIC POWER

In general, power is defined as the rate of doing work or rate of spending energy. Similarly, the electric power is defined as the rate of consumption of electrical energy. It represents the rate at which the electrical energy is converted into some other form of energy. The SI unit of electric power is watt

Resistor	Variable resistor or Rheostat	Voltmeter	Galvano meter	A diode	Potentiom eter	Emitting Diode
•Used to fix the magnitude of the current through a circuit	•Used to select the magnitude of the current through a circuit	•Used to measure the potential difference	• Used to indicate the direction of current.	• A diode has various uses, which you will study in higher classes	•The Potentiom eter is an instrument used for the measurem ent of potential difference	•A LED has various uses.

ELECTRONICS

Electronics is that field of science which deals with the motion of electrons under the influence of applied electric and/or magnetic field. Electronics can be classified into two branches:

Physical Electronics and Electronics Engineering

Semiconductor Materials

In general, semiconductors are special class of elements having a conductivity between that of a good conductor and that of an insulator

Intrinsic Semiconductor
An extremely pure semiconductor is called intrinsic semiconductor

Extrinsic Semiconductor
Although an intrinsic semiconductor is capable to conduct a little current even at room temperature but as it is, it is not useful for the preparation of various electronic devices. To make it conductive a small amount of suitable impurity is added. It is then called extrinsic (impure) semiconductor

Doping

The process by which an impurity is added to a semiconductor is known as doping

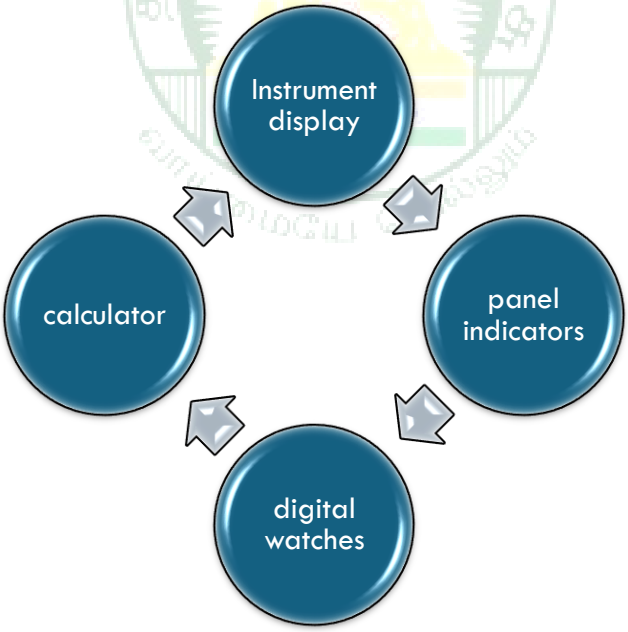
- Depending upon the type of impurity added extrinsic semiconductor may be classified as

1. n- type semiconductor
2. p-type semiconductor

LED (Light Emitting Diode)

When a diode is forward biased the potential barrier is lowered. The conduction band free electrons from n- region cross the barrier and enter the p-region, as these electrons enter the p- region they fall into the holes lying in the valence band. Hence they fall from a higher energy level to a lower energy level in the process they radiate energy. The LED are different. These are made of gallium arsenide phosphide (GaAsP) and gallium phosphide (GaP). In LED the energy is radiated in the form of light and hence they glow. A manufacturer can produce LED that radiate red, green, yellow, blue, orange light.

Application



HEAT, LIGHT AND SOUND

HEAT

- Heat is an energy that raises the temperature of a thing by causing the molecules in that thing to move faster.
- SI Unit of Heat is joule. The unit calorie is also used.

Transfer of heat

- Heat transfer takes place in three ways

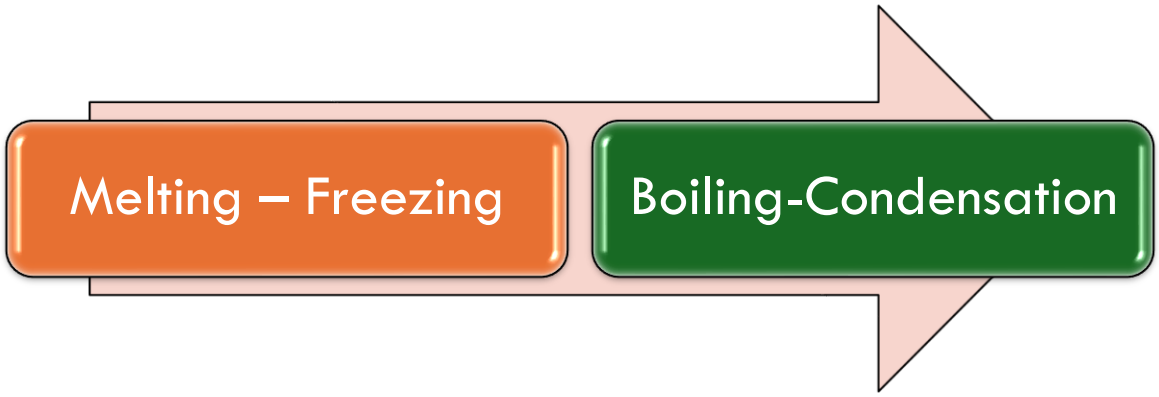
Conduction	Convection	Radiation
• The process of transfer of heat in solids from a region of higher temperature to a region of lower temperature without the actual movement of molecules is called conduction. • e.g. Mercury is used in thermometers because it is a good conductor of heat.	• Convection is the flow of heat through a fluid from places of higher temperature to places of lower temperature by movement of the fluid itself. • eg. Sea breeze and Land Breeze	• Radiation is the flow of heat from one place to another by means of electromagnetic waves. • e.g. Base of cooking utensils is blackened because black surface absorb more heat from the surrounding

Sublimation.

- The process in which a solid is converted to gaseous state is called sublimation.

Change of state

- The process of changing of a substance from one physical state to another at a definite temperature is defined as change of state.



LIGHT

- Light is a form of energy which travels in the form of waves. The path of light is called ray of light and group of these rays are called as beam of light.

PROPERTIES OF LIGHT

- Light is a form of energy.
- Light always travels along a straight line.
- Light does not need any medium for its propagation. It can even travel through vacuum.
- The speed of light in vacuum or air is, $c = 3 \times 108$ ms-1.
- Since, light is in the form of waves, it is characterized by a wavelength (λ) and a frequency (ν), which are related by the following equation: $c = \nu \lambda$ (c - velocity of light).
- Different coloured light has different wavelength and frequency.
- Among the visible light, violet light has the lowest wavelength and red light has the highest wavelength.
- When light is incident on the interface between two media, it is partly reflected and partly refracted.

SCATTERING OF LIGHT

When sunlight enters the Earth's atmosphere, the atoms and molecules of different gases present in the atmosphere refract the light in all possible directions. This is called as 'Scattering of light'.

Types of scattering

When a beam of light, interacts with a constituent particle of the medium, it undergoes many kinds of scattering. Based on initial and final energy of the light beam, scattering can be classified as,

1)Elastic scattering 2) Inelastic scattering

Elastic scattering If the energy of the incident beam of light and the scattered beam of light are same, then it is called as 'elastic scattering'

Inelastic scattering If the energy of the incident beam of light and the scattered beam of light are not same, then it is called as 'inelastic scattering'. The nature and size of the scatterer results in different types of scattering. They are Rayleigh scattering , Mie scattering Tyndall scattering, Raman scattering.

Rayleigh scattering

- The scattering of sunlight by the atoms or molecules of the gases in the earth's atmosphere is known as Rayleigh scattering.
- 'S' $\propto 1/\lambda 4$

Mie scattering

- Mie scattering is caused by pollen, dust, smoke, water droplets, and other particles in the lower portion of the atmosphere.
- Mie scattering is responsible for the white appearance of the clouds

Tyndall Effect

- The scattering of light rays by the colloidal particles in the colloidal solution is called Tyndall Scattering or Tyndall Effect

Raman Scattering

- Raman Scattering is defined as "The interaction of light ray with the particles of pure liquids or transparent solids, which leads to a change in wavelength or frequency." The spectral lines having frequency equal to the incident ray frequency is called 'Rayleigh line' and the spectral lines which are having frequencies other than the incident ray frequency are called 'Raman lines'

LENSES

- A lens is an optically transparent medium bounded by two spherical refracting surfaces or one plane and one spherical surface. Lens is basically classified into two types.
- They are: (i) Convex Lens (ii) Concave Lens.

Convex or bi-convex lens: It is a lens bounded by two spherical surfaces such that it is thicker at the centre than at the edges. A beam of light passing through it, is converged to a point. So, a convex lens is also called as converging lens

Concave or bi-concave Lens: It is a lens bounded by two spherical surfaces such that it is thinner at the centre than at the edges. A parallel beam of light passing through it, is diverged or spread out. So, a concave lens is also called as diverging lens

Other types of Lenses

Plano-convex lens: If one of the faces of a bi-convex lens is plane, it is known as a plano-convex lens.

Plano-concave lens: If one of the faces of a bi-concave lens is plane, it is known as a plano-concave lens.

Lens Formula :

- The lens formula gives the relationship among distance of the object (u), distance of the image (v) and the focal length (f) of the lens.
- It is expressed as $1/f = 1/v - 1/u$

Differences between a Convex Lens and a Concave Lens

S.No	Convex Lens	Concave Lens
1	A convex lens is thicker in the middle than at edges.	**A concave lens is thinner in the middle than at edges.**
2	It is a converging lens.	**It is a diverging lens.**
3	It produces mostly real images.	**It produces virtual images.**
4	**It is used to treat hypermeteropia.**	**It is used to treat myopia.**

Myopia	• Myopia, also known as short sightedness
Hypermeteropia	• Hypermeteropia, also known as long sightedness
Presbyopia	•Due to ageing, ciliary muscles become weak and the eye-lens become rigid (infl exible) and so the eye loses its power of accommodation. Because of this, an aged person cannot see the nearby objects clearly. So, it is also called as 'old age hypermetropia

Telescope

- Telescope is an optical instrument to see the distant objects. The first telescope was invented by Johann Lippershey in 1608.

Types of Telescope

Refracting Telescopes :

In refracting telescope lenses are used. Galilean telescope, Keplerian telescope, Achromatic refractors are some refracting telescopes.

Reflecting Telescopes :

In reflecting telescope parabolic mirrors are used. Gregorian, Newtonian. Cassearain telescope are some reflecting telescopes.

SOUND

- Sound is a form of energy which produces sensation of hearing in our ears. Human ear can hear only a particular range of frequency of sound that too with a certain range of energy.

Properties

- Sound needs a medium for Propagation
- A sound wave is an example of a longitudinal mechanical wave.
- Sound can travel only when there are particles which can be compressed and rarefied.
- Compressions are the regions where particles are crowded together.
- Rarefactions are the regions of low pressure where particles are spread apart.

Characteristics of a sound wave

Amplitude (A)	Frequency (n)	Time period (T)	Wavelength (λ)	Velocity or speed (v)
• The maximum displacement of the particles of the medium from their original undisturbed positions, when a wave passes through the medium is called amplitude of the wave. • Amplitude is denoted as A. Its SI unit is meter (m)	• The number of vibrations (complete waves or cycles) produced in one second is called frequency of the wave. It is denoted as n. The SI unit of frequency is hertz (Hz).	• The time required to produce one complete vibration (wave or cycle) is called time period of the wave. It is denoted as T. The SI unit of time period is second (s)	• The minimum distance in which a sound wave repeats itself is called its wavelength. The wavelength is usually denoted as λ (Greek letter lambda). The SI unit of wavelength is metre (m).	• The distance travelled by the sound wave in one second is called velocity of the sound. The SI unit of velocity of sound is m s–1

Laws of reflection

The angle of incidence ∠i is equal to the angle of reflection ∠r

APPLICATIONS REFLECTION OF SOUND

- Sound board
- Ear trumpet
- Mega phone

DOPPLER EFFECT

- When ever there is a relative motion between a source and a listener, the frequency of the sound heard by the listener is different from the original frequency of sound emitted by the source. This is known as "Doppler effect".

Applications of Doppler effect

(a) To measure the speed of an automobile
(b) Tracking a satellite
(c) RADAR (RAdio Detection And Ranging)
(d) SONAR
(e) Ultrasonic sound or Ultrasound
(f) Hearing Aid